

Problem Statement No.4: Sequential Task Agent – Investment Analyst

The Challenge

Investment analysts need to process multiple layers of information—financial reports, stock performance data, market news, and analyst opinions. Much of this information is fragmented and requires a sequence of tasks (data gathering, validation, analysis, comparison, and reporting) to generate useful insights. Manual workflows are time-consuming and prone to errors. The challenge is not just to collect market data but to automate sequential tasks that lead to actionable investment recommendations.

The Objective

Build a **Sequential Task Agent** that acts as an intelligent investment analyst. The solution should:

1. **Automate Data Gathering:** Collect financial data from multiple trusted sources such as company reports, stock exchanges, and market news feeds.
2. **Perform Multi-Step Analysis:** Sequentially process data (e.g., extract KPIs → calculate financial ratios → benchmark against competitors → analyze market sentiment).
3. **Generate Predictive Insights:** Identify trends, detect risks, and provide forward-looking recommendations (e.g., *“Stock X shows declining margins—consider underweighting in portfolio”*).
4. **Deliver Structured Reports:** Provide an easy-to-understand dashboard with investment summaries, trend charts, risk indicators, and AI-generated recommendations.

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models